JIJI GOMEZ

AI Engineer

Kerala, India • +91 8921423584 • jijigomez89@gmail.com • GitHub: jijigomez • LinkedIn: LinkedIn.com/in/jiji-gomez

PROFESSIONAL SUMMARY

Results-oriented AI Engineer with 3+ years of experience designing and deploying production ready machine learning and generative AI solutions. Proven expertise in building scalable AI applications, optimizing model performance, and integrating leading-edge LLMs and RAG systems. Proficient in Python, TensorFlow, Hugging Face and cloud-based deployment on AWS. Proven ability to deliver AI-powered solutions that optimize efficiency and create business value.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Pandas, NumPy, Data Preprocessing, Data Cleaning and analysis

Machine Learning: Scikit-learn, Supervised/Unsupervised Learning, Predictive Modeling, Feature Engineering, Model Optimization, Hyperparameter Tuning

Deep Learning: TensorFlow, Keras, PyTorch, CNN, Transfer Learning, Data Augmentation

Generative AI & LLMs: Large Language Models, Retrieval-Augmented Generation (RAG), Prompt Engineering, Hugging Face Transformers, LangChain, Vector Embeddings, Semantic Search

NLP & Computer Vision: Text Classification, Named Entity Recognition, Image Classification, Object Detection, Image Segmentation, OCR, Transformer Models

Web & API Development: Django, FastAPI, Flask, Streamlit, React, REST API Design, Microservices

Cloud & Deployment: AWS (EC2, S3, Lambda), Docker, CI/CD Pipelines, Model Serving, MLOps

Edge & Embedded AI: Real-time Audio Processing, Signal Processing, Hardware Integration, Edge Device Optimization

Data Visualization & BI: Power BI, Tableau, Matplotlib, Seaborn, Dashboard Development

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code

PROFESSIONAL EXPERIENCE

AI Engineer | Spectrum Softech Solutions Pvt Ltd, Kochi

August 2024 – Present

- Built and deployed 8+ AI and ML solutions in production, covering NLP, computer vision and recommendation systems.
- Built LLM and RAG applications that let enterprise clients search their own documents and get answers in context.
- Improved inference accuracy by 20%+ and cut latency by 15% through feature engineering and model tuning.
- Exposed models as real-time prediction services with FastAPI and Django; the APIs handle 10,000+ requests a day at 99.5% uptime.
- Deployed on AWS (EC2, S3) with Docker, and set up CI/CD pipelines for automated testing and releases.
- Worked with product, data and engineering teams to turn business requirements into working AI features.

Junior Data Scientist | Oracuz Infotech Pvt Ltd, Thiruvananthapuram

December 2023 – August 2024

- Built 6+ machine learning models (classification, regression, clustering) and tracked their performance against business goals.
- Automated data pipelines in Python for 50GB+ datasets, cutting manual processing time by 30%, about 20 hours a month.
- Built Power BI and Tableau dashboards so stakeholders could follow key metrics without waiting on reports.
- Tuned models with hyperparameter search and ensembles, improving operational efficiency by 15% and reducing forecast errors by 22%.

KEY PROJECTS

Real-Time Speech Enhancement on Edge Hardware(Edge AI)

- Built a real-time audio system on the Femto SPU-001 EVK2 that reduces background noise by 85% while keeping speech clear.
- Handled the full pipeline: firmware optimization, model deployment and hardware integration. Tested with live radio broadcast audio.
- Reached inference latency under 50 ms, meeting real-time requirements.
- Tech: Embedded AI, edge computing, Python, real-time audio and signal processing

Retrieval-Augmented Generation (RAG) Question-Answering Platform

- Built a document Q&A system that pairs vector embeddings with LLMs to give accurate, context-aware answers.
- Implemented semantic chunking, embedding generation and vector similarity search, retrieving relevant context with 95%+ accuracy.
- Used LangChain and Hugging Face Transformers to handle complex questions that span multiple documents.
- Tech: RAG, LLMs, Python, LangChain, Hugging Face, vector databases

Computer Vision: Image Classification & Object Detection

- Trained CNN classifiers with 94%+ accuracy using transfer learning (ResNet, VGG) and data augmentation.
- Built object detection pipelines on pre-trained models, with REST APIs for batch and real-time inference.
- Added image segmentation and OCR for processing structured documents.
- Tech: TensorFlow, Keras, PyTorch, OpenCV, REST APIs

EDUCATION

Master of Science (M.Sc.) in Electronics • Marthoma College of Science and Technology, Kollam

June 2022

CERTIFICATIONS & TRAINING

• Data Analytics Professional Certificate, Cisco (2023)

LANGUAGES

English (Fluent), Malayalam (Native)